

$R\vec{\xi} = 0$
not parametrizable ⚡

Idea: Lift system into higher dimension where it is parametrizable

add 1 or 2 ghost task(s)

ξ_{GT}

Λ renders system
projective or
torsion free

$$(R \quad -\Lambda) \begin{pmatrix} \vec{\xi} \\ \xi_{GT} \end{pmatrix} = 0$$

condition $\xi_{GT} = 0$

minimally parametrized
(Ghost tasking)

$$R\vec{\xi} = 0 \Leftrightarrow \mathcal{L}\vec{u} = \vec{f}$$

add (many) inhomogeneity tasks

$f_1, \dots, f$

$\mathbb{I}_q$ renders system
projective

$$(\mathcal{L} \quad -\mathbb{I}_q) \begin{pmatrix} \vec{\xi} \\ f_1 \\ \dots \\ f_q \end{pmatrix} = 0$$

condition $f_1, \dots, f_q$

over parametrized
standard physics-informed
Gaussian Processes